



INSuRE Fall 2023 Midterm Presentation Malware Similarity

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Team

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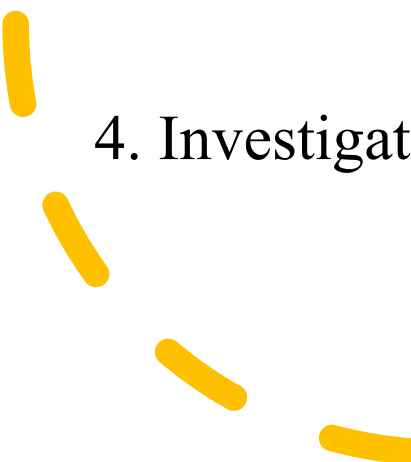


Description

- Our project, guided by MITRE's Technical Director Frank Posluszny, delves into the fascinating realm of cybersecurity.
 - We're on a mission to connect professionals working on similar malware projects across the country. By identifying the unique fingerprints left by the Malware Development Life Cycle (MDLC), we aim to foster collaboration and reduce duplication of effort.
 - Through this project, we envision a future where cybersecurity experts come together seamlessly, enhancing our collective ability to safeguard digital landscapes.
 - This initiative not only bolsters our understanding of cyber threats but also propels us towards a more secure and interconnected digital world.
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Motivation

1. Analyzing malware behavior for pattern recognition.
 2. Evaluating the effectiveness of current anti-malware solutions.
 3. Implementing proactive measures to safeguard against evolving threats.
 4. Investigating novel approaches for malware identification and mitigation.
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Goals

Uncovering Malware Relationships:

1. Investigating the connections between different malware strains.
2. Tracing the evolution of malware families and their variants.

Behavioral Profiling of Malware:

1. Analyzing the actions and interactions of various malware types.
2. Identifying patterns in their behavior for predictive analysis.

Enhancing Threat Intelligence:

1. Developing insights that aid in early detection and response to emerging malware threats.
2. Enabling a proactive approach to cybersecurity through in-depth understanding.

Goals (Continued)

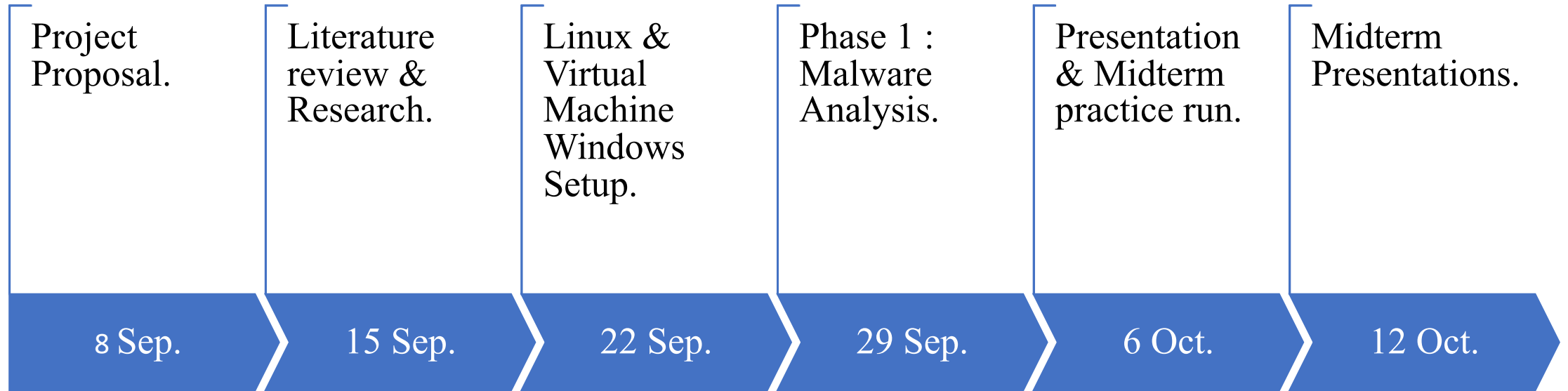
Advancing Cybersecurity Collaboration:

1. Creating a platform for experts to share insights and collectively combat malware.
2. Fostering a community-driven approach to enhance global digital security.

Improving Security Operations:

1. Providing actionable intelligence to Security Operations Centers (SOCs) for effective response.
2. Enabling SOC analysts to make informed decisions based on comprehensive malware analysis.

Timelines



Literature Review

- 1. Malware Similarity Analysis: An adversary perspective - Frank Posluszny, March 2020
- 2. Software Forensics: Research Paper - Eugene H. Spaffordl and Stephen A. WeebeT
- 3. Automatic analysis of malware behavior using machine learning: Research Article - [Rieck,](#) [Konrad,](#) [Trinius, Philipp,](#) [Willems, Carsten,](#) [Holz,](#) [Thorsten](#)

Malware Similarity Analysis - An adversary perspective

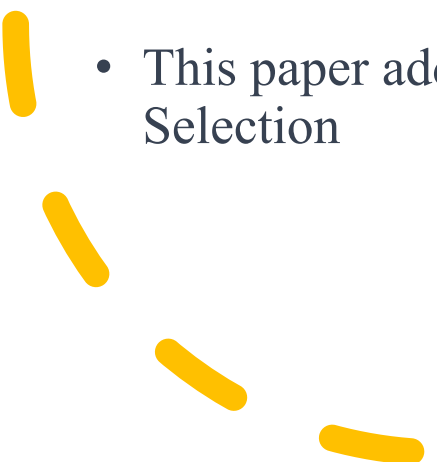
- The article presents the Malware Development Life Cycle (MDLC), mirroring the Software Development Life Cycle with eight phases. It discusses each stage and its relevance to malware analysis, emphasizing MDLC's utility in identifying similarities among malware instances, aiding practitioners in understanding adversaries' tactics.
- Challenges: Malware Diversity, Data Collection, Integration into Existing Frameworks, Measurement of Artifacts
- This paper addresses these challenges by Malware Similarity Identification, Data Integration, Improved Defenses, Further Research



Figure 1 : Malware Development Life Cycle (simplified)



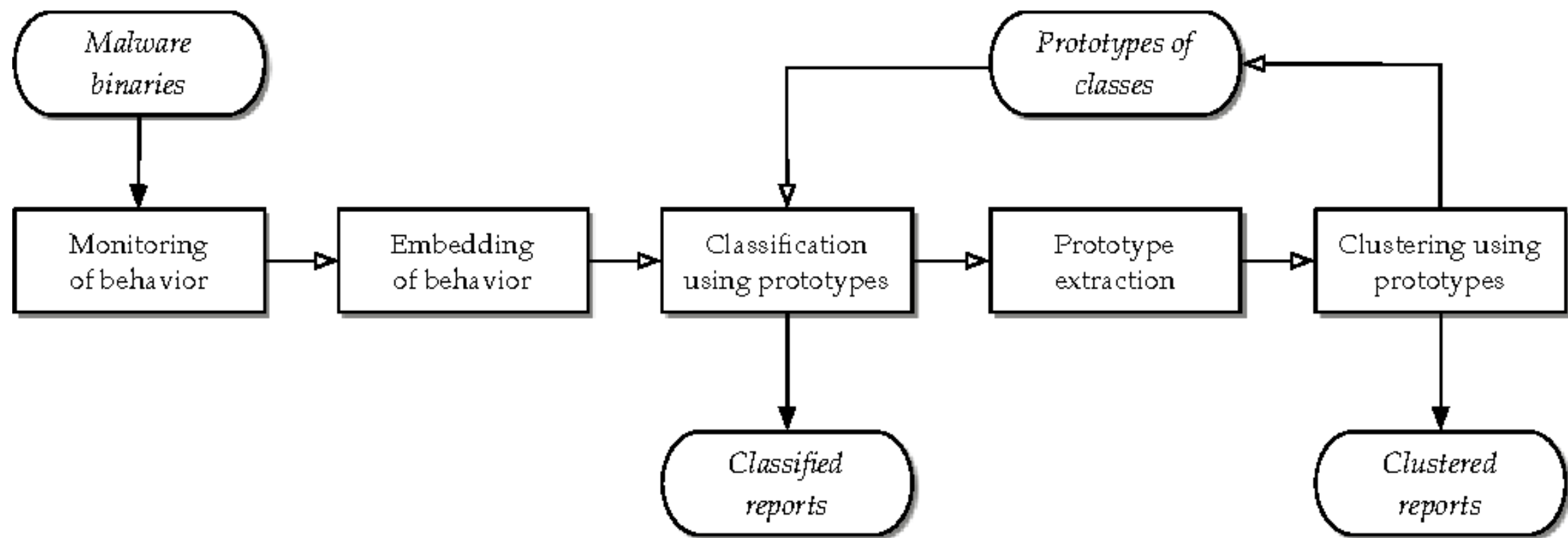
Software Forensics – Can we track code to its authors?

- The paper emphasizes that hackers and virus authors rely on network anonymity, making it hard to catch them. To address this, it recommends reducing anonymity through techniques like detailed session logging and enhanced network protocols with user identification.
 - Challenges : Anonymity in Cybercrimes, Effectiveness of Countermeasures, Authorship Identification ,Sample Selection
 - This paper addresses these challenges by Reducing Anonymity, Authorship Identification , Sample Selection
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Automatic analysis of malware behavior using machine learning

- The paper deals with the persistent problem of malware in computer security. It introduces a machine learning system to analyze malware behavior, enabling the detection of new types and categorizing unknown malware. Additionally, it offers an efficient method for this analysis that maintains accuracy while reducing runtime demands.
 - Challenges : Malware Diversity, Real-Time Analysis, Clustering and Classification , Runtime Overhead.
 - The challenges are addressed by Machine Learning, Incremental Analysis, Behavior-Based Analysis, Accurate Discovery
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Challenges



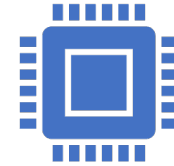
1. Less number of teammates in our group.



2. Limited and varying domain specific knowledge among team members.



3. Despite a large amount of available data, there is currently a need for deeper analysis to derive meaningful insights.



4. Windows and Linux Virtual environment support on Windows and M1 Mac Machines.

Next Steps



SHORT TERM GOAL 1 :



SOFTWARE INSTALLATION



SHORT TERM GOAL 2:



DEVELOP A SCRIPT TO
OVERCOME THE PROBLEM
BY TAKING ADVANTAGE OF
LITERATURE SURVEY
PAPERS.



DOING DEEP ANALYSIS.



LONG TERM GOAL:



CODE IMPLEMENTATION



CREATING REPORTS.



Any Questions?





THANK YOU!!

